

Math 432/532: Introduction to Topology II

Solutions to HW #9

1. Consider the discrete persistence module

$$\mathbb{R}^2 \xrightarrow{f} \mathbb{R}^3 \xrightarrow{g} \mathbb{R}^2,$$

where f is represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ and g by the matrix $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(i) Compute the persistent Betti numbers $\beta_{1,2}$, $\beta_{1,3}$, and $\beta_{2,3}$ for this discrete persistence module.

Solution: The first map has rank 2, so $\beta_{1,2} = 2$. The second map has rank 2, so $\beta_{2,3} = 2$. The composition of the two maps (product of the two matrices) has rank 1, so $\beta_{1,3} = 1$.

(ii) Draw a persistence diagram to represent this discrete persistence module. (This amounts to computing the numbers $n_{s,t}$, where $s < t \in \{1, 2, 3, \infty\}$.)

Solution: Lemma 9.2.2 gives us the following equations:

$$\beta_{1,3} = n_{1,\infty}, \quad \beta_{1,2} = n_{1,3} + n_{1,\infty}, \quad \beta_{2,3} = n_{1,\infty} + n_{2,\infty}.$$

From these equations and part (i), we conclude that $n_{1,\infty} = 1$, $n_{1,3} = 1$, and $n_{2,\infty} = 1$. That is, the persistence diagram consists of three dots, in locations $(1, \infty)$, $(1, 3)$, and $(2, \infty)$.

(iii) Draw a barcode to represent this discrete persistence module.

Solution: From part (ii), we see that the barcode consists of one bar from 1 to ∞ , one bar from 1 to 3, and one bar from 2 to ∞ .

2. Recall Problem 1(iii) from Homework #8. We take $X = S^1 \subset \mathbb{R}^2$ with the Euclidean metric, let $S = \{(1, 0), (0, 1), (-1, 0), (0, -1)\} \subset X$, and consider the Čech filtration $\{\text{Cech}(S, r) \mid r \geq 0\}$. Let $V(i)_r = H_i(\text{Cech}(S, r))$, with maps $h(i)_{r,s} : V(i)_r \rightarrow V(i)_s$ induced by the inclusion $\text{Cech}(S, r) \subset \text{Cech}(S, s)$ for all $r \leq s$. Draw the barcodes for $V(0)$, $V(1)$, and $V(2)$.

Solution: On the previous homework, we saw that the realization of $\text{Cech}(S, r)$ consists of four points when $r < \sqrt{2} - \sqrt{2}$, it is homeomorphic to a circle when $r \in [\sqrt{2} - \sqrt{2}, \sqrt{2})$, it is homeomorphic to S^2 when $r \in [\sqrt{2}, \sqrt{2} + \sqrt{2})$, and it is homeomorphic to a 3-dimensional ball when $r > \sqrt{2} + \sqrt{2}$. We know the homology of all of these complexes!

- The module $V(0)$ consists of four copies of $\mathbb{R}$ that are born at scale 0 and die and collapse to a single copy of $\mathbb{R}$ at scale $\sqrt{2} - \sqrt{2}$, and then remain a single copy of $\mathbb{R}$ for all time. Thus the barcode for $V(0)$ consists of one copy of the bar $[0, \infty)$ and three copies of the bar $[0, \sqrt{2} - \sqrt{2})$.
- The module $V(1)$ consists of one copy of $\mathbb{R}$ that is born at scale $\sqrt{2} - \sqrt{2}$ and dies at scale $\sqrt{2}$, so its barcode consists of the single bar $[\sqrt{2} - \sqrt{2}, \sqrt{2})$.
- The module $V(2)$ consists of one copy of $\mathbb{R}$ that is born at scale $\sqrt{2}$ and dies at scale 2 , so its barcode consists of the single bar $[\sqrt{2}, \sqrt{2} + \sqrt{2})$.

3. This problem is the missing piece needed to complete our proof that the interleaving distance between two interval modules is equal to the bottleneck distance between the corresponding persistence diagrams. Suppose that we have real numbers $p < q$, $p' < q'$, and $\epsilon \geq 0$. Furthermore, suppose that the both of the following conditions hold:

$$(a) \ p + \epsilon \geq p' \text{ and } q - \epsilon \leq q',$$

$$(d) \ p' + \epsilon \geq q' - \epsilon.$$

Prove that at least one of the following two conditions hold:

$$(c) \ p' + \epsilon \geq p \text{ and } q' - \epsilon \leq q,$$

$$(b) \ p + \epsilon \geq q - \epsilon.$$

Solution: Let's assume that (a) and (d) hold, and that (b) fails (i.e. $p + \epsilon < q - \epsilon$). We need to show that (c) holds. We have

$$q' - \epsilon \leq p' + \epsilon \leq p + 2\epsilon < q,$$

so the second part of (c) holds. Similarly,

$$p' + \epsilon \geq q' - \epsilon \geq q - 2\epsilon > p,$$

so the first part of (c) holds.